

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to:

- _____
- _____
- _____

Vocabulary & Key Concepts

implicit curve:

parametrization:

ellipse parametrization:

parabola parametrization:

Hook

The ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ is *not* a function — it fails the vertical line test. But if a particle traces this ellipse over time, we can use parametric tools to find its position, extremes, and intercepts. How do we convert this implicit equation to a parametric form?

Section 1 — Ellipse Parametrization

Setup. Starting from $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$, set $\frac{x-h}{a} = \cos t$ and $\frac{y-k}{b} = \sin t$.

Then: $x(t) = \underline{\hspace{2cm}}$ and $y(t) = \underline{\hspace{2cm}}$

Domain: $0 \leq t \leq \underline{\hspace{2cm}}$

Verify. Substitute back into the ellipse equation and use the identity $\cos^2 t + \sin^2 t = 1$:

$$\frac{(x(t)-h)^2}{a^2} + \frac{(y(t)-k)^2}{b^2} = \frac{(\underline{\hspace{2cm}})^2}{a^2} + \frac{(\underline{\hspace{2cm}})^2}{b^2} = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \quad \checkmark$$

Example. Ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$. Here $h = \underline{\hspace{1cm}}$, $k = \underline{\hspace{1cm}}$, $a = \underline{\hspace{1cm}}$, $b = \underline{\hspace{1cm}}$.

Parametrization: $x(t) = \underline{\hspace{2cm}}$ $y(t) = \underline{\hspace{2cm}}$ $0 \leq t \leq \underline{\hspace{2cm}}$

Maximum $x(t)$: $\underline{\hspace{2cm}}$ when $t = \underline{\hspace{2cm}}$ $\Rightarrow$ Point: $\underline{\hspace{2cm}}$

Extrema from the parametrization. Minimum $x(t)$: $\underline{\hspace{2cm}}$ when $t = \underline{\hspace{2cm}}$ $\Rightarrow$ Point: $\underline{\hspace{2cm}}$

Maximum $y(t)$: $\underline{\hspace{2cm}}$ when $t = \underline{\hspace{2cm}}$ $\Rightarrow$ Point: $\underline{\hspace{2cm}}$

Section 2 — Parabola Parametrization

Setup. For $y - k = a(x - h)^2$ (vertical parabola): set $x(t) = \underline{\hspace{2cm}}$, then $y(t) = \underline{\hspace{2cm}}$, $t \in \mathbb{R}$.

Vertex at $t = 0$: $\underline{\hspace{2cm}}$

Example. $y = x^2 - 4x + 3$.

Step 1 — Complete the square: $y = (x - \underline{\hspace{1cm}})^2 - \underline{\hspace{1cm}}$. Vertex: $(\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$.

Step 2 — Parametrize: $x(t) = \underline{\hspace{2cm}}$, $y(t) = \underline{\hspace{2cm}}$.

Step 3 — Fill the table:

t	$x(t) = 2 + t$	$y(t) = t^2 - 1$	Point
-2	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$
-1	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$
0	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$
1	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$
2	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$	$\underline{\hspace{2cm}}$

Section 3 — Applying Parametric Tools to Implicit Curves

Ellipse: $x(t) = 3 \cos t$, $y(t) = 5 \sin t$.

- When does the curve cross the **y -axis**? Set $x(t) = 0$: $\cos t = 0 \Rightarrow t = \underline{\hspace{2cm}}$
Coordinates at those t -values: $\underline{\hspace{2cm}}$
- When does the curve cross the **x -axis**? Set $y(t) = 0$: $\sin t = 0 \Rightarrow t = \underline{\hspace{2cm}}$
Coordinates at those t -values: $\underline{\hspace{2cm}}$
- What is the **rightmost** point? Max of $x(t) = 3 \cos t$ is $\underline{\hspace{1cm}}$, occurring at $t = \underline{\hspace{1cm}}$. Point: $\underline{\hspace{2cm}}$
- What is the **topmost** point? Max of $y(t) = 5 \sin t$ is $\underline{\hspace{1cm}}$, occurring at $t = \underline{\hspace{1cm}}$. Point: $\underline{\hspace{2cm}}$